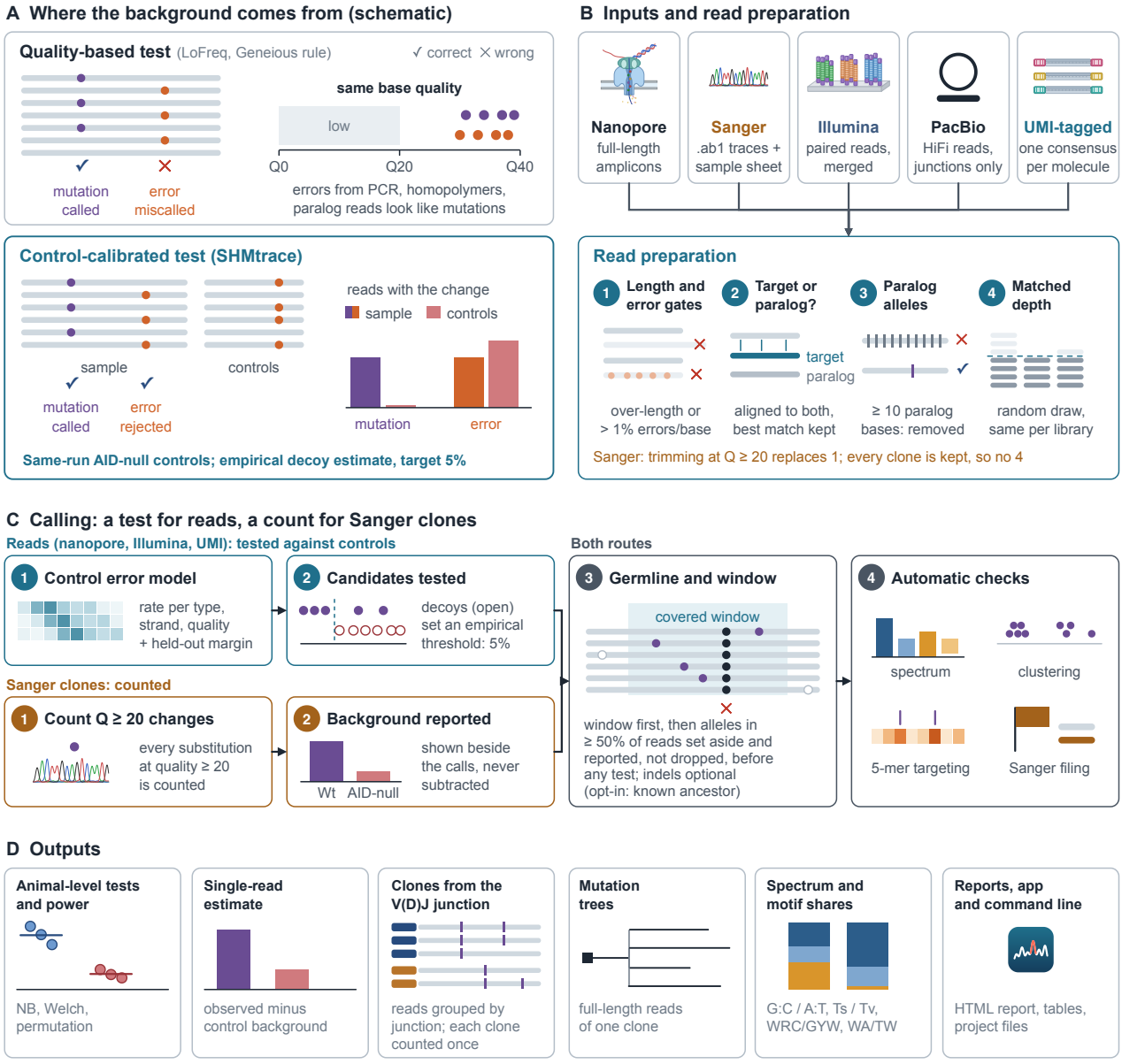

A Where the background comes from (schematic)
Quality-based test (LoFreq, Geneious rule)
✓ correct ✕ wrong
mutation called
error miscalled
same base quality
low
Q0
Q20
Q40
errors from PCR, homopolymers, paralog reads look like mutations
Control-calibrated test (SHMtrace)
sample
controls
mutation called
error rejected
reads with the change
sample
controls
mutation
error
Same-run AID-null controls; empirical decoy estimate, target 5%
B Inputs and read preparation
Nanopore
full-length amplicons
Sanger
.ab1 traces + sample sheet
Illumina
paired reads, merged
PacBio
HiFi reads, junctions only
UMI-tagged
one consensus per molecule
Read preparation
1 Length and error gates
2 Target or paralog?
target
paralog
3 Paralog alleles
4 Matched depth
over-length or > 1% errors/base
aligned to both, best match kept
≥ 10 paralog bases: removed
random draw, same per library
Sanger: trimming at Q ≥ 20 replaces 1; every clone is kept, so no 4
C Calling: a test for reads, a count for Sanger clones
Reads (nanopore, Illumina, UMI): tested against controls
1 Control error model
rate per type, strand, quality + held-out margin
2 Candidates tested
decoys (open) set an empirical threshold: 5%
Sanger clones: counted
1 Count Q ≥ 20 changes
every substitution at quality ≥ 20 is counted
2 Background reported
shown beside the calls, never subtracted
Wt
AID-null
Both routes
3 Germline and window
covered window
window first, then alleles in ≥ 50% of reads set aside and reported, not dropped, before any test; indels optional (opt-in: known ancestor)
4 Automatic checks
spectrum
clustering
5-mer targeting
Sanger filing
D Outputs
Animal-level tests and power
NB, Welch, permutation
Single-read estimate
observed minus control background
Clones from the V(D)J junction
reads grouped by junction; each clone counted once
Mutation trees
full-length reads of one clone
Spectrum and motif shares
G:C / A:T, Ts / Tv, WRC/GYW, WA/TW
Reports, app and command line
HTML report, tables, project files